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Alpilla 600 MW Combined Cycle Power Plant

Secondary Fuel: Light Diesel Oil (LDO)

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Function	Owner's Engineer (discipline lead)	Owner's Engineer (project engineer)	Owner's Project Director
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CASE STUDY - all organisations, data and signatures are fictional.

1. Purpose and operating philosophy

Light diesel oil is the secondary (back-up) fuel, used when natural gas is curtailed by the network operator or unavailable.

- Expected use: up to 500 hours per year (emergency back-up fuel status under the environmental permit), typically in winter.
- Autonomy: **72 hours of base-load operation** on LDO from full storage.
- Automatic transfer from gas to LDO on loss of gas pressure, without unit trip; manual transfer back to gas on load.
- Start-up on LDO shall be possible (gas not available).

2. Fuel quality as delivered (supplier's specification, based on TS EN 590 automotive diesel, class B/F)

Property	Unit	Limit / value	Test method
Density at 15 °C	kg/m ³	820–845	EN ISO 12185
Kinematic viscosity at 40 °C	mm ² /s	2.0–4.5	EN ISO 3104
Flash point	°C	≥ 55	EN ISO 2719
Cloud point / CFPP (winter grade)	°C	≤ -5 / ≤ -15	EN 23015 / EN 116
Sulphur	mg/kg	≤ 10	EN ISO 20846
Water	mg/kg	≤ 200	EN ISO 12937
Total contamination (particulates)	mg/kg	≤ 24	EN 12662
Ash	% m/m	≤ 0.01	EN ISO 6245
Carbon residue (10 % distillation residue)	% m/m	≤ 0.30	EN ISO 10370
FAME (biodiesel) content	% v/v	≤ 7.0	EN 14078
Net calorific value (LHV)	MJ/kg	42.6–43.2 (typical 42.9)	ASTM D240
Sodium + potassium	mg/kg	≤ 1.0 (as delivered, not guaranteed by supplier)	ASTM D3605 / ICP
Vanadium	mg/kg	≤ 0.1	ASTM D5708
Lead	mg/kg	≤ 0.5	ICP
Calcium	mg/kg	≤ 1.0	ICP
Nitrogen (fuel-bound)	mg/kg	≤ 150	ASTM D4629

The supplier does not guarantee alkali metal content after road transport. Past deliveries to other plants showed Na+K up to 1.8 mg/kg after contamination with water in tanker trucks.

3. Supply logistics

- Delivery by road tankers of 30–36 m³ from the refinery terminal (95 km).
- The supplier can guarantee **up to 25 trucks per day** in winter emergencies.
- The Owner will purchase the first fill of the LDO tanks; the Contractor provides LDO for its own testing beyond the first fill.

4. Owner requirements

- Truck unloading station with at least 4 bays, metering, filtration and spill containment.
- Storage: two tanks, each 50 % of the 72-hour autonomy, in a bunded area with foam fire protection (NFPA 11 / NFPA 30).
- Water drain, sampling and fuel quality monitoring. Fuel treatment if needed to meet the gas turbine OEM limits at the burner.
- Emissions on LDO: as per the environmental permit (see ALP-OWN-ENV-001).